ModelIt!

Student Activity Pack

# Can We Reverse Aging?

Modeling Telomere Dynamics, Cellular Senescence, and Epigenetic Reprogramming

Grade 12L3 | NGSS HS-LS1-4 | Stability and Change

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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# 1 Pre-Assessment

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**What Do You Know?** Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **What do you think causes aging at the cellular level? Why do old cells work differently than young cells?**

2. **If scientists could slow or reverse the aging process, do you think they should? What are the potential benefits and risks?**

3. **Why do you think some people seem biologically younger or older than their chronological age? What might explain the difference?**

4. **What is the difference between extending lifespan (living longer) and extending healthspan (living healthier)? Which matters more?**

## 2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

| Skill | Expert (4) | Proficient (3) | Developing (2) | Beginning (1) |
|---|---|---|---|---|
| **Identifying Components** | EXPERT<br>Correctly identifies and sorts ALL components as internal or external with detailed reasoning | PROFICIENT<br>Correctly identifies and sorts MOST components with some reasoning | DEVELOPING<br>Identifies components but has difficulty sorting or explaining | BEGINNING<br>Needs significant help identifying components |
| **Establishing Relationships** | EXPERT<br>Accurately describes ALL positive and negative relationships with clear evidence | PROFICIENT<br>Accurately describes MOST relationships with some evidence | DEVELOPING<br>Identifies some relationships but struggles with direction | BEGINNING<br>Needs help understanding how components connect |
| **Running Simulations** | EXPERT<br>Makes accurate predictions AND explains surprising results using evidence | PROFICIENT<br>Makes reasonable predictions and records clear observations | DEVELOPING<br>Runs simulations but predictions or observations are incomplete | BEGINNING<br>Needs help running simulations or recording data |
| **Explaining Results** | EXPERT<br>Uses multiple pieces of evidence to explain HOW and WHY systems interact | PROFICIENT<br>Uses evidence to explain system interactions | DEVELOPING<br>Gives partial explanations with limited evidence | BEGINNING<br>Needs help connecting evidence to explanations |

# 3 Component Analysis

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### What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

### Sort These Components

- Chronological Age
- Oxidative Stress Level
- Caloric Intake Level
- Telomere Length
- Senescent Cell Burden
- Chronic Inflammation Level
- Stem Cell Function
- Epigenetic Age
- Mitochondrial Function
- Healthspan Index

| EXTERNAL | INTERNAL |
|---|---|
| | |

### Explain your reasoning: Why did you sort each component the way you did?

# 4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

**? Explain ONE relationship in your model. What happens when one component changes?**

# 5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

## 1 Scenario 1: Normal Aging

Chronological Age: 20->80 | Oxidative Stress: Moderate | Caloric Intake: Normal

MY PREDICTION

WHAT I OBSERVED

## 2 Scenario 2: Caloric Restriction

30% caloric restriction starting at age 40 | All other inputs unchanged

MY PREDICTION

WHAT I OBSERVED

## 3 Scenario 3: Senolytic Therapy

Normal aging to age 65 | Then senolytic treatment reducing senescent cells by 50%

MY PREDICTION

WHAT I OBSERVED

# 6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

## Possible New Components

**Autophagy Rate** — The efficiency of the cell's self-cleaning system that digests and recycles damaged proteins, dysfunctional mitochondria, and other cellular debris — autophagy declines with age, allowing damaged components to accumulate, and activating autophagy through fasting or rapamycin is one of the most promising anti-aging strategies

**NAD+ Level** — The cellular concentration of nicotinamide adenine dinucleotide, a critical coenzyme for energy metabolism and DNA repair — NAD+ declines by approximately 50% between ages 40 and 60, and restoring NAD+ levels with precursors like NMN or NR has improved healthspan markers in animal studies

**Immune Senescence** — The progressive decline of immune system function with age — fewer naive T cells, accumulated memory cells, reduced vaccine response, and impaired cancer surveillance that increases susceptibility to infections and malignancy in older adults

## My Extension

**A Which new component would you add to the model?**

**B How would it connect to the existing components? (Draw or describe)**

**C What evidence or reasoning supports your connection?**

# 7 Reflection Questions

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### The Big Question

**Every cell in your body carries a molecular clock that counts down with each division. When the clock runs out, the cell stops dividing, begins to malfunction, and eventually poisons its neighbors. This process — cellular aging — drives wrinkles, organ failure, cancer, and death. Is aging an inevitable law of biology, or a disease that could be treated and even reversed?** Use evidence from your simulation to support your answers.

1. **How does your model demonstrate why aging accelerates after a certain age rather than progressing at a constant rate?**

2. **What does the senolytic therapy scenario reveal about whether aging is irreversible damage or actively maintained decline?**

3. **Why is caloric restriction effective at slowing aging according to your model, and what does this suggest about the role of metabolic rate in longevity?**

4. **How would adding autophagy or NAD+ levels to your model change the predicted effectiveness of anti-aging interventions?**

5. **What are the ethical implications of anti-aging therapies that could be very expensive — should longevity be available only to those who can afford it?**

# 8 STEM Challenge: Design an Anti-Aging Clinical Trial Protocol

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## YOUR ENGINEERING MISSION

Design a clinical trial protocol testing a multi-intervention anti-aging strategy, specifying which aging mechanisms each intervention targets, how you would measure biological age changes, and what ethical considerations apply to longevity research.

### THE SCENARIO

A longevity research company has developed a combination anti-aging therapy: a senolytic drug to clear senescent cells, an NAD+ precursor to restore mitochondrial function, and a low-dose rapamycin analog to activate autophagy. Your team must design the Phase 2 clinical trial: determine the target population (age range, health status), specify which biomarkers of biological age you will measure (epigenetic clock, telomere length, inflammatory markers, functional tests), define primary and secondary endpoints, and address the ethical challenges of testing a therapy designed not to treat disease but to slow the aging process itself.

### Engineering Design Thinking

1. WHAT IS YOUR TARGET POPULATION AND WHAT INCLUSION/EXCLUSION CRITERIA ENSURE A MEANINGFUL TEST OF THE INTERVENTION?

2. WHICH BIOMARKERS WILL YOU MEASURE AND HOW DO THEY CONNECT TO THE AGING MECHANISMS IN YOUR MODEL?

3. WHAT IS THE MINIMUM TRIAL DURATION NEEDED TO DETECT MEANINGFUL CHANGES IN BIOLOGICAL AGING RATE?

4. HOW DOES YOUR TRIAL ADDRESS THE POSSIBILITY THAT SLOWING AGING FOR SOME COULD WIDEN HEALTH DISPARITIES IF THE THERAPY IS EXPENSIVE?

### My Design

DESIGN AREA — DRAW YOUR SOLUTION

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**! Explain your design using scientific evidence from the simulation:**

### Real Career Connection

Gerontologists who study the biology of aging earn $70,000-$140,000/year at research universities and biotech companies. Longevity medicine physicians who apply anti-aging science to clinical practice earn $150,000-$350,000/year. Biotech researchers developing senolytic drugs and epigenetic therapies at companies like Unity Biotechnology and Altos Labs earn $100,000-$200,000/year. Bioethicists specializing in longevity and life extension earn $65,000-$130,000/year.